Sahrdaya College of Engineering and Technology, Kodakara

Answer Key

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

SECOND SEMESTER B.TECH DEGREE EXAMINATION, OCTOBER 2021

Course Code: EST130

Course Name: BASICS OF ELECTRICAL AND ELECTRONICS ENGINEERING

PART II: BASIC ELECTRIONICS ENGINEERING

PART A

Answer all questions. Each question carries 4 marks

Q13 **Differentiate between Avalanche breakdown and Zener breakdown? (4)**

ans

Zener Breakdown	Avalanche breakdown
1.This occurs at junctions which being heavily doped have narrow depletion layers	1. This occurs at junctions which being lightly doped have wide depletion layers.
2. This breakdown voltage sets a very strong electric field across this narrow layer.	2. Here electric field is not strong enough to produce Zener breakdown.
3. Here electric field is very strong to rupture the covalent bonds thereby generating electron-hole pairs. So even a small increase in reverse voltage is capable of producing Large number of current carriers.	3. Her minority carriers collide with semi conductor atoms in the depletion region, which breaks the covalent bonds and electron-hole pairs are generated. Newly generated charge carriers are accelerated by the electric field which results in more collision and generates avalanche of charge carriers. This results in avalanche breakdown.
4. Zener diode exhibits negative temp: coefficient. Ie. breakdown voltage decreases as temperature increases.	4. Avalanche diodes exhibits positive temp: coefficient. i.e breakdown voltage increases with increase in temperature.

Q14 **Draw and explain the block diagram of a public address system. (4)**

ans P.A (Public Address) system, or sound reinforcement system, is a collection of equipment designed to amplify voice and music so that they can be heard by a larger number of people. From the small speaker that makes the announcement at school to the several clock loads of equipment used by big touring acts, the principle main the same:

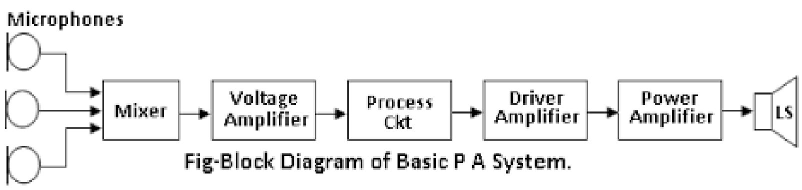
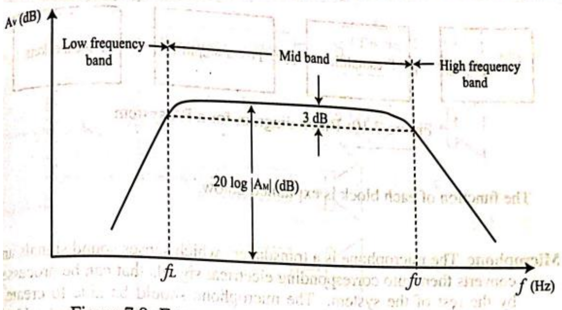


Fig-Block Diagram of Basic P A System.

1. The Original sound travels to one or more microphones.
2. The Microphones convert the sound to electrical impulses and send it down another wire to a mixing console.

		3. The Mixing console mixes all the signal together in a correct balance and sends this single 'mixed' signal down another wire to an amplifier. 4. The Amplifier converts the electrical signal into much larger ones and sends them down another wire to teh speaker(s) 5. The Speaker convert the electrical into mechanical vibration, setting up sound waves. 6. The Audience hears these sound waves.
	Q15	**Give reasons for decrease in transistor amplifier gain at low frequencies and high frequencies. (4)**
	ans	Figure 7.9 shows the frequency response of RC-coupled amplifier. In ac coupled amplifiers, coupling capacitors or transformers are used to pass the signal from one stage to another. The presence of these coupling capacitors reduces the gain at low frequencies. At low frequencies, capacitive impedance increases as it is inversely proportional to frequency of the signal ($X_C = \frac{1}{2\pi fC}$). The gain is also reduced at high frequencies due to the presence of internal capacitances such as stray capacitance. These capacitances are parasitic capacitances internally formed at transistor junctions. The gain will be constant over a mid-frequency range. All coupling and by-pass capacitors are considered short-circuit at midband frequencies while all internal capacitive effects are considered open-circuit.  Figure 7.9: Frequency response of RC-coupled amplifier
	Q16	**Explain the relevance of Intermediate Frequency in a superheterodyne receiver. (4)**
	ans	In communications and electronic engineering, an **intermediate frequency** (**IF**) is a frequency to which a carrier wave is shifted as an intermediate step in transmission or reception.The intermediate frequency i created by mixing the carrier signal with a local oscillator signal in a process called heterodyning, resulting in a signal at the difference or beat frequency. Intermediate frequencies are used in superheterodyne radi receivers, in which an incoming signal is shifted to an IF for amplification before final detection is done. Conversion to an intermediate frequency is useful for several reasons. When several stages of filters are used, they can all be set to a fixed frequency, which makes them easier to build and to tune. Lower frequency transistors generally have higher gains so fewer stages are required. It's easier to make sharply selective filters at lower fixed frequencies.

PART B

Answer any one full question from each module. Each question carries 10 marks

Module 4

Q17a **17 a) What are the different types of inductors? Give two typical applications of inductor. (5)**

ans

Air Core Inductor

Ceramic core inductors are referred as “Air core inductors”. Ceramic is the most commonly used materia for inductor cores. Ceramic has very low thermal co-efficient of expansion, so even for a range of operatin temperatures the stability of the inductor’s inductance is high. Since ceramic has no magnetic propertie: there is no increase in the permeability value due to the core material.Its main aim is to give a form for th coil. In some cases it will also provide the structure to hold the terminals in place. The main advantage c these inductors are very low core losses, high Quality factor. These are mainly used in high frequenc applications where low inductance values are required

Iron Core Inductor

In the areas where low space inductors are in need then these iron core inductors are best option.Thes inductors have high power and high inductance value but limited in high frequency capacity. These ar applicable in audio equipments. When compared with other core indictors these have very limite applications.

Ferrite Core Inductor

Ferrite is also referred as ferromagnetic material. They exhibit magnetic properties. They consist of mixe metal oxide of iron and other elements to form crystalline structures . The general composition of ferrite is XFe2O4. Where X represents transition materials. Mostly easily magnetized material combinations ar used such as manganese and zinc (MnZn), nickel and zinc (niZn).Ferrites are mainly two types they are sot ferrites and hard ferrites. These are classified according to the magnetic coercivity. Coercivity is th magnetic field intensity needed to demagnetize the ferromagnetic material from complete saturation stat to zero.

Soft Ferrite:These materials will have the ability to reverse their polarity of their magnetization without an particular amount of energy needed to reverse the magnetic polarity.

Hard Ferrite:**These are also called as permanent magnets. These will keep the polarity of th**

magnetization even after removing the magnetic field.

Ferrite core inductor will help to improve the performance of the inductor by increasing the permeabilit of the coil which leads to increase the value of the inductance. The level of the permeability of the ferrit core used within the inductors will depend on the ferrite material. This permeability level ranges from 2 to 15,000 according to the material of ferrite. Thus the inductance is very high with ferrite core whe compared to the inductor with air core.

Iron Powder Inductor

These are formed from very fine particles with insulated particles of highly pure iron powder. This type c inductor contains nearly 100% iron only. It gives us a solid looking core when this iron power is compresse under very high pressure and mixed with a binder such as epoxy or phenolic. By this action iron powde forms like a magnetic solid structure which consists of distributed air gap.Due to this air gap it is capable t store high magnetic flux when compared with the ferrite core. This characteristic allows a higher D current level to flow through the inductor before inductor saturates. This leads to reduce the permeabilit of the core.Mostly the initial permeability’s are below 100 only. Thus these inductors posses with hig temperature co-efficient stability. These are mainly applicable in switching power supplies.

applications of inductors are:

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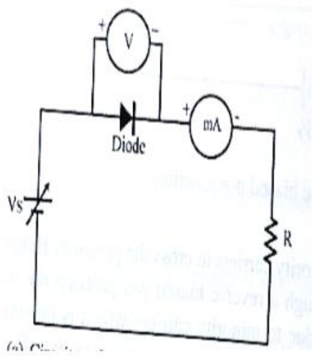
- Choking, blocking, attenuating, or filtering/smoothing high frequency noise in electrical circuits.
- Storing and transferring energy in power converters (dc-dc or ac-dc)
- Creating tuned oscillators or LC (inductor / capacitor) "tank" circuits.
- Impedance matching.

</td></tr>
<tr><td></td><td>b</td><td>Describe the VI characteristics of PN junction diode. (5)</td></tr>
<tr><td></td><td>ans</td><td>

(i) Forward characteristics



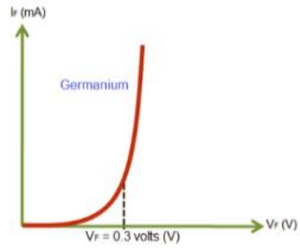


Fig: Forward characteristics of germanium diode

The positive terminal of the battery repels majority carriers, holes, in P-region and negative terminal repels electrons in the N region and pushes

them towards the junction. This result in increase in concentration of charge carriers near junction, recombination takes place and width of depletion region decreases. As forward bias voltage is raised depletion region continues to reduce in width, and more and more carriers recombine. This results in an exponential rise of current.

Cut-in or Knee voltage :

The voltage at which the current starts to conduct rapidly in a forward biased p-n junction diode.

(Vk of Si=0.7V)(Vk of Ge= 0.3V)

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(ii) Reverse characteristics

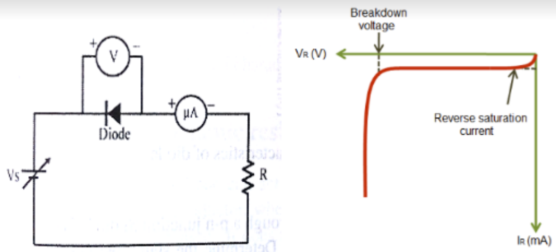

Fig: Reverse characteristics of diode

Negative terminal of the battery attracts majority carriers, holes, in P-region and positive terminal attracts electrons in the N-region and pull them away from the junction. This result in decrease in concentration of charge carriers near junction and width of depletion region increases.

A small amount of current flow due to minority carriers, called as reverse bias current or leakage current. As reverse bias voltage is raised depletion region continues to increase in width.

Q18 a

Derive the relation between common base current gain and common emitter current gain,(4)

ans

Formulae used:

The current gain α in Common Base CB mode is given by

$$\alpha = \frac{I_c}{I_e} \text{ ...... (1)}$$

Here, I_c is the current from the collector and I_e is the current form emitter.

The current gain β in Common Emitter CE mode is given by

$$\beta = \frac{I_c}{I_b} \text{ ...... (2)}$$

Here, I_c is the current from the collector and is the current form base. I_b

		We have asked to determine the relationship between the current gain α in Common Base CB mode and the current gain β in Common Emitter CE mode.From equation (1). We can write $\frac{1}{\alpha}=\frac{I_e}{I_c}$ From equation (2). We can write $\frac{1}{\beta}=\frac{I_b}{I_c}$ We know that in a transistor the sum of the current from the collector and the current from base is equal to the current from the emitter. $I_e=I_c+I_b$ Multiply both sides of the above equation by the collector current I_c. $\frac{I_e}{I_c}=1+\frac{I_b}{I_c}$ Substitute $\frac{1}{\alpha}$ for $\frac{I_e}{I_c}$ and $\frac{1}{\beta}$ for $\frac{I_b}{I_c}$ in the above equation. $\frac{1}{\alpha}=1+\frac{1}{\beta}$ $\Rightarrow\frac{1}{\beta}=\frac{1}{\alpha}-1$ $\Rightarrow\frac{1}{\beta}=\frac{1-\alpha}{\alpha}$ $\therefore\beta=\frac{\alpha}{1-\alpha}$ Therefore, the relationship between the current gain α in Common Base CB mode and the current gain β in Common Emitter CE mode is $\beta=\frac{\alpha}{1-\alpha}$.
	b	**Sketch the output characteristic of a transistor and explain different regions of operation. (6)**

ans

CE output characteristics

For obtaining common emitter output characteristics, we set base current to a constant value (say $I_B = 40\ \mu A$) by adjusting base-to-emitter voltage. Then the collector-to-emitter voltage (V_{CE}) is varied by changing the power supply voltage and corresponding collector currents are noted. If we plot a graph with V_{CE} along x-axis and I_C along y-axis, we obtain a curve as shown in Figure 5.8. This procedure is repeated for different values of base current resulting in similar curves.

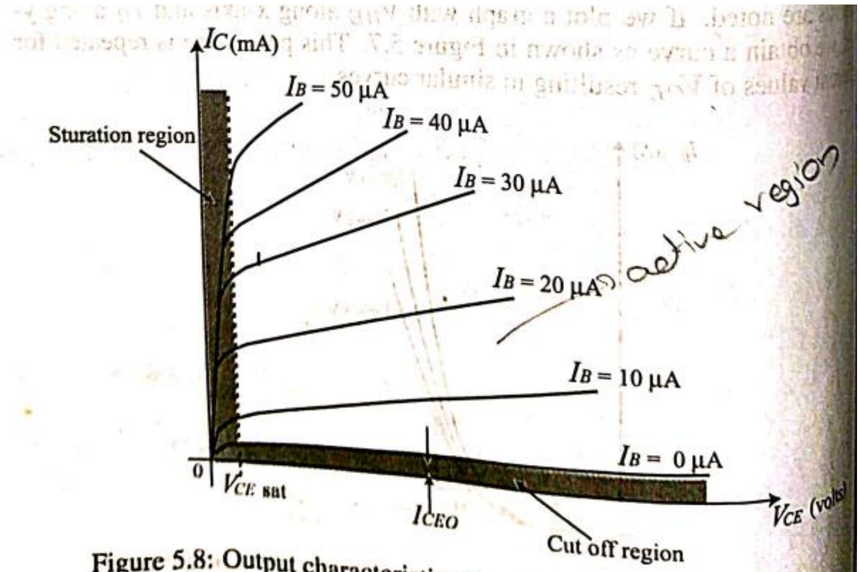

Figure 5.8: Output characteristics for transistor in CE configuration

The graph consists of three important regions; saturation, cut-off and

		For increased values of V_{CE}, I_C increases rapidly to a saturation value, depending upon the value of base current. When V_{CE} is raised further, I_C slightly increases. This occurs due to the decrease in the base current for increased value of V_{CE}. (This happens due to Early effect, which is explained in the next section). When base current is zero, a small collector current which is known as leakage current I_{CEO} flow from collector to emitter. The region below $I_B = 0$ is known as cut-off region. In this region both emitter junction and collector junction are reverse biased. In the active region, emitter junction is forward biased and collector junction is reverse-biased. Transistor is operated within the active region when it is used as amplifier. It operates in saturation and cut-off for switching applications. Output resistance and current gain are determined from the output characteristics. Output resistance is the ratio of a change in collector-to-emitter voltage to resulting change in collector current for a constant base current.
Module 5		
	Q19 a	Explain the working of a full wave bridge rectifier. (5)
	ans	The working & operation of a full wave bridge rectifier is pretty simple. The circuit diagrams and waveforms we have given below will help you understand the operation of a bridge rectifier perfectly. In the circuit diagram, 4 diodes are arranged in the form of a bridge. The transformer secondary is connected to two diametrically opposite points of the bridge at points A & C. The load resistance R_L is connected to bridge through points B and D. ***During the first half cycle*** During the first half cycle of the input voltage, the upper end of the transformer secondary winding is positive with respect to the lower end. Thus during the first half cycle diodes D1 and D3 are forward biased and current flows through arm AB, enters the load resistance R_L, and returns back flowing through arm DC. During this half of each input cycle, the diodes D2 and D4 are reverse biased and current is not allowed to flow in arms AD and BC During the second half cycle During the second half cycle of the input voltage, the lower end of the transformer secondary winding is positive with respect to the upper end. Thus diodes D2 and D4 become forward biased and current flows through arm CB, enters the load resistance R_L, and returns back to the source flowing through arm DA. Thus the direction of flow of current through the load resistance R_L remains the same during both half cycles of the input supply voltage.

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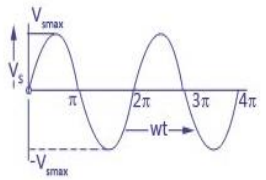
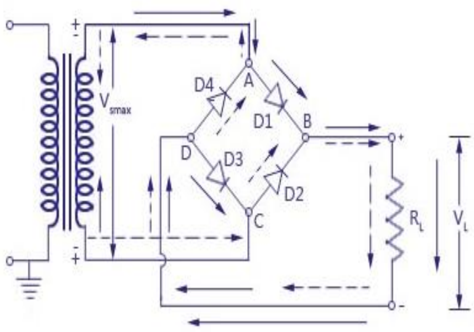
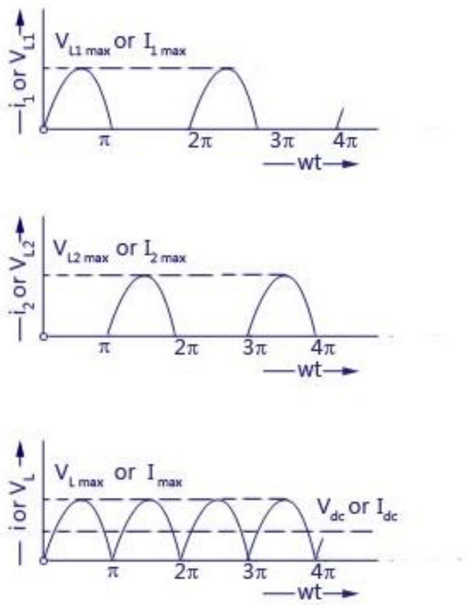


Input Voltage Waveform





Rectified Output Voltage/Current Waveforms
</td></tr>
<tr><td></td><td>b</td><td>Explain the working of an RC coupled amplifier. (5)</td></tr>
<tr><td></td><td>ans</td><td>Common Emitter (CE) amplifier is widely used in audio frequency applications in radio and TV receivers. It provides current, voltage and power gains. For the proper functioning of an amplifier, the transistor must be biased in the active region where the base current has complete control over the collector current. Thus a small increase in the base current results in a relatively large increase in the collector current and a small decrease in the base current is followed by a large decrease in the collector current.
When large voltage gain is required, many stages of amplifiers are used. The coupling between successive stages are done using different methods namely; resistance-coupling, transformer coupling and direct coupling. Circuit shown in Figure 7.8 is RC-coupled common emitter amplifier. The coupling stage in RC-coupled amplifier forms an RC network. Voltage divider bias technique is</td></tr>
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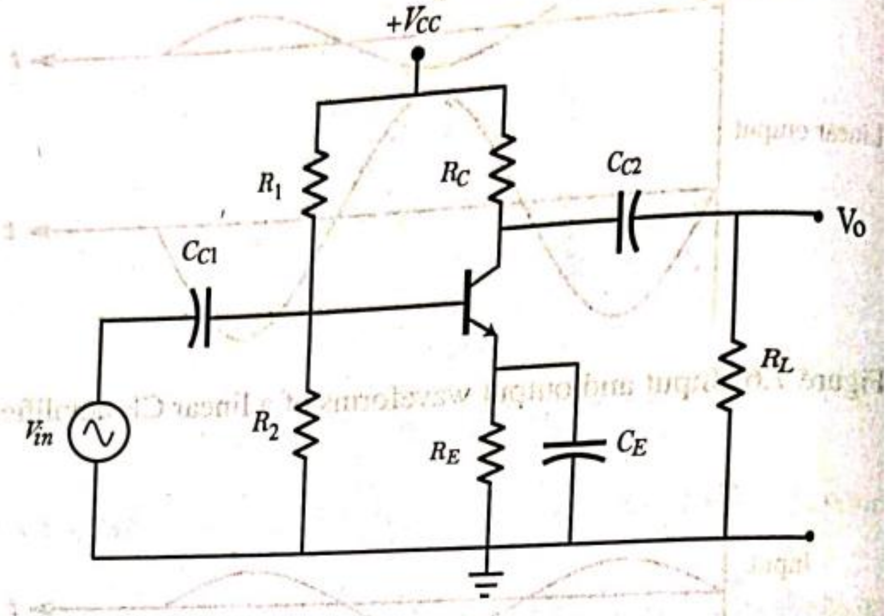
		employed in this circuit.  Figure 7.8: RC-coupled amplifier Resistors R_1 and R_2 are called potential divider resistors because they develop certain fixed voltage across them, derived from V_{CC} supply. Resistance R_E stabilizes the operating point against temperature variation. The purpose of the coupling capacitors is to couple the ac signal and block dc. Coupling capacitor C_{C1} couples the input ac signal V_{in} to the input of the amplifier. Coupling capacitor C_{C2} is used to block the dc component of the output voltage from reaching the load resistor R_L. The purpose of the bypass capacitor C_E is to bypass the ac signal developed across R_E, to ground. Presence of C_E raises the gain of the amplifier but reduces its bandwidth.
	Q20 a	**Describe the working of a zener diode voltage regulator. (5)**
	ans	A zener diode provides *constant voltage* when operated in the *reverse breakdown region* irrespective of current through it and variations in the applied voltage. Thus a zener diode can be used as voltage regulator to provide constant dc output voltage. Figure 6.16 shows the circuit of a zener diode voltage regulator. Since the zener diode is connected in parallel with the load, the circuit is also known as shunt regulator. The resistance R_S is known as series current limiting resistor which is used to limit current in the circuit. The output voltage is taken across the load resis-

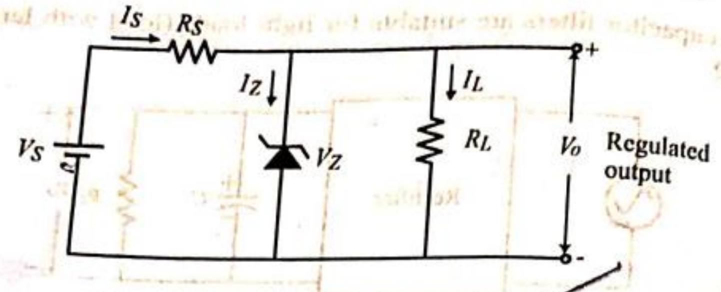


Figure 6.16: Zener diode voltage regulator

$V_S > V_Z$

tance R_L. It is essential that the input voltage V_S must be greater than the zener voltage so that the zener operates in the zener breakdown region.

Design of zener voltage regulator

Following points are considered while designing a shunt zener voltage regulator.

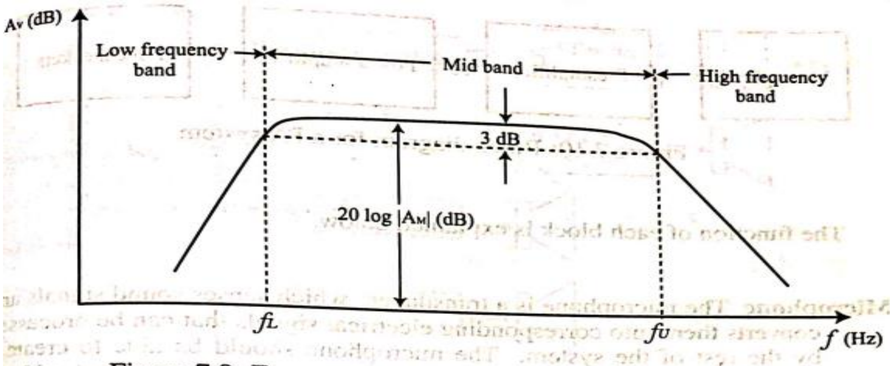
1. For acting as a voltage regulator, zener diode must be operated in the breakdown region or regulating region. Zener current should be maintained between $I_{Z\min}$ and $I_{Z\max}$. The current $I_{Z\min}$ is the minimum zener current required to maintain the zener diode in the ON state i.e., regulating region. The current $I_{Z\max}$ is the maximum zener current that zener diode can conduct without being damaged.
2. The zener diode should be selected in such a way that its maximum power rating is not exceeded. If maximum power dissipation of a zener is $P_{Z\max}$ and zener voltage is V_Z, then, $P_{Z\max} = V_Z I_{Z\max}$

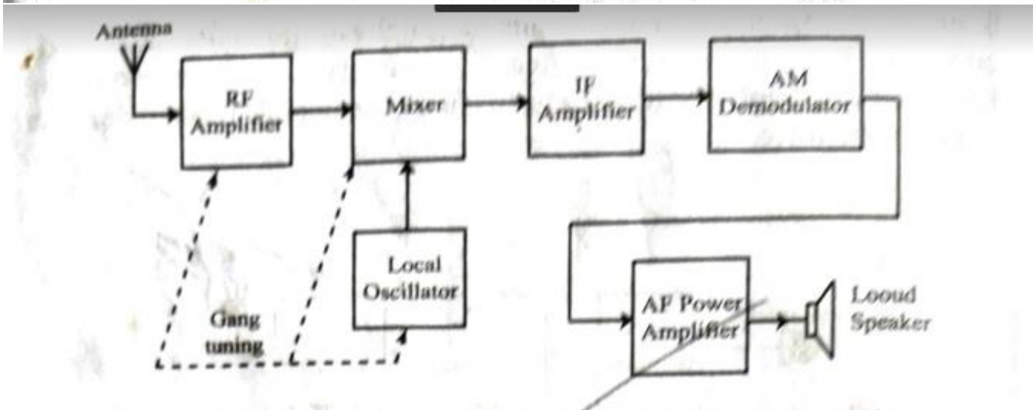
$$I_{Z\max} = \frac{P_{Z\max}}{V_Z}$$

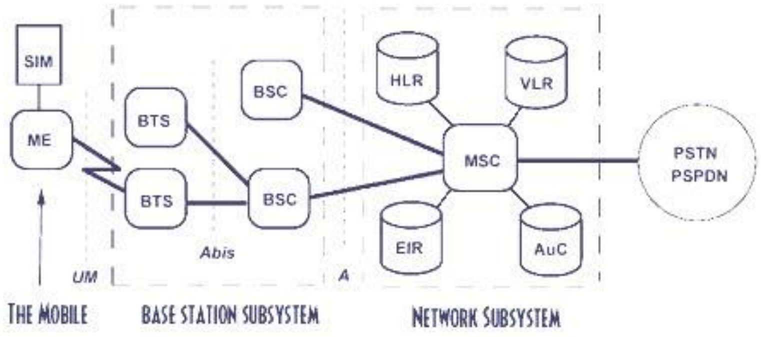
3. The minimum value of the applied input voltage $V_{S\min}$ must be higher than V_Z, preferably at least twenty-five percent higher.
4. There should be a minimum value of load resistance, $R_{L\min}$ to ensure that zener diode will remain in the regulating region.

$$I_S = \frac{V_S - V_o}{R_S}$$

	b	**Draw and explain the frequency response of an RC coupled amplifier. (5)**

	ans	The essential purpose of an amplifier is to accept an input signal and provide an amplified version of that signal as an output. Every amplifier has a frequency-response curve associated with it and it provides information about the performance of the amplifier at various frequencies. The frequency response of an amplifier is a graph of frequency (usually in x-axis) and gain (usually in y-axis). No real amplifier has equal gain at all frequencies. For an amplifier, there will be *mid-band gain*, that is, the gain of the amplifier in the middle of its normal operating frequency range. In this range, the gain will be normally constant. The frequencies at which the gain falls by 3 dB from the mid-band gain are termed as *lower cut-off frequency* (f_L) and *upper cut-off frequency* (f_U).  Figure 7.9: Frequency response of RC-coupled amplifier Figure 7.9 shows the frequency response of RC-coupled amplifier. In ac coupled amplifiers, coupling capacitors or transformers are used to pass the signal from one stage to another. The presence of these coupling capacitors reduces the gain at low frequencies. At low frequencies, capacitive impedance increases as it is inversely proportional to frequency of the signal ($X_C = \frac{1}{2\pi fC}$). The gain is also reduced at high frequencies due to the presence of internal capacitances such as stray capacitance. These capacitances are parasitic capacitances internally formed at transistor junctions. The gain will be constant over a mid-frequency range. All coupling and by-pass capacitors are considered short-circuit at midband frequencies while all internal capacitive effects are considered open-circuit.
		Module 6
	Q21	**With a neat sketch explain AM super heterodyne receiver. (5)**

	ans	Heterodyne means mixing. The block diagram of an AM super heterodyne receiver is shown in Figure 11.10. The antenna picks up the modulated signals of several stations which are within the receiving range of the radio receiver. The RF amplifier is tuned to the required incoming frequency and so it selects only the desired signal while rejecting all other signals. RF amplifier also amplifies the incoming week signal before it is applied to the mixer. In the mixer circuit, the incoming signal frequency is converted to a new frequency by mixing it with another signal generated by a local oscillator.  Figure 11.10: Block diagram for AM Super heterodyne receiver The new frequency will be the difference between the incoming signal frequency and local oscillator frequency and it is called the intermediate frequency (IF). This frequency is standardized as 455 kHz for all AM broadcast receivers. The IF signal from the mixer circuit is then amplified by the IF amplifier. It is then fed to the detector stage. The detector separates audio component from the modulated signal. Thus the original audio signal is extracted. The audio frequency signal is amplified by the power amplifier to raise the power level in order to drive the loudspeaker. The loud speaker converts the audio signal into sound wave of corresponding frequency. Superheterodyne receiver has many advantages such as better adjacent channel rejection, high sensitivity and selectivity etc.
	Q22	**Describe the principle and block diagram of a GSM system. (5)**
	ans	Fig 1 shows GSM architecture. The network manly consist of User Equipment (UE), Base transceiver station (BTS), Mobile switching center (MSC). The GSM contains most of the necessary capabilities to support packet transmission over GSM. The critical part in the GPRS network is the mobile to GSN (MS-SGSN) link which includes the MS-BTS, BTS-BSC, BSC-SGSN, and the SGSN-GGSN link .fig 1 shows block diagram of GSM architecture.



User Equipment (UE)—These are the users .Number of users are controlled by one BTS
1. The mobile stations (MS) communicate with the base station subsystem over the radio the radio interface.
2. The BSS called as radio the subsystem, provides and manages the radio transmission path between the mobile stations and the Mobile Switching Centre(MSC).It also manages radio interface between the mobile stations and other subsystems of GSM.
3. Each BSS comprises many Base Station Controllers(BSC) that connect the mobile station to the network and switching subsystem (NSS) through the mobile switching cente
4. The NSS controls the switching functions of the GSM system.It allows the mobile switching center to communicate with networks like PSTN, ISDN, CSPDN, PSPDN and other data networks.
5. The operation support system (OSS) allows the operation and mantanance of the GSM system. It allows the system engineers to diagnose,troubleshoot and observe the parameters of the GSM systems.The OSS subsystem interacts with the other subsystems and is provided for the GSM operating company staff that provides service facilities for the network.

Base station(BSS)-- The following stations subsystem comprises of two parts:
1. Base Transceiver Station (BTS).
2. Base Station Controller(BSC).

The BSS consists many BSC that connect to a single MSC. Each BSC controls upto several hundred BTS.

Base Transreciver Station(BTS)-BTS

It has radio transreciever that define a cell and are capable of handelling radio link protocols with MS.

Functions of BTS are
1. Handelling radio link protocols
2. Providing FD communication to MS.
3. Interliving and de- interliving.

Base station controller(BSC) IT manages radio resources for one or more BTS.It controls several hundred BTS al are connected to single MSC.

Functions of BTS are
• To control BTS.
• Radio resource management
• Handoff management and control
• Radio channel setup and frequency hoping

Network subsystem(NSS)

1.It handels the switching of GSM calls between external networks and indoor BSC
2.It includes three different data bases for mobility management as

		A .HLR (Home Location Register) B .VLR (Visitor Location Register) C. AUC (Authentication center) **Mobile switching center (MSC)--** It connects fix networks like ISDN ,PSTN etc. Following are the functions of MSC 1. Call setup, supervision and relies 2. COLLECTION OF BILLING INFORMATION 3. Call handelling / routing 4. Management of signelling protocol 5. Record of VLR and HLR **HLR (Home Location Register)** - Call roamiming and call routing capabilities of GSM are handeled.It stores all the adminstartive information of sub scriber registered in the networks.IT maintance unique international mobile subscriber identity.(IMSI). **VLR (Visitor Location Register)** - It is a temporary data base.It stores the IMSC number and costomer information for each roaming customer visiting specific MSC. **Authentication center** - It is protected database .It maintance authentication keys and algorithms.It contain s aregister called as Equipment Identity Register. **Operation subsystem(OSS)** - IT manages all mobile equipment in the system 1)management for charging and billing procedure 2)To maintain all hardware and network operations
	b	**Explain the concept of cells and frequency reuse in cellular communication. (5)**
	ans	**Frequency Reuse:** It is the process in which the same set of frequencies can be allocated to more than one cell,provide the cells are at a certain minimum distance apart referred to as frequency reuse distance.The design process of selecting and allocating channels groups for all of the cellular base station within a system is called Frequency Reuse or Frequency Planning. **Co-channel interference:** Cell which uses the same set of frequencies are referred to as co-channel cells.In the above figure co channel cells are labelled with same name such as A,B,C - - - - F. The space between adjacent channels is filled with other cells that use different frequencies to provide frequency isolation.If the system is not properly designed co-channel interference may occurs due to the simultaneous use of the same channel.Specifically if the available channels are reused for additional traffic,it is possible to serve more number of users.Frequency channels are allocated to each cell by means of an intelligent frequency planning technique so as to minimize the co-channel and adjacent channel interference while meeting the performance requirements both in terms of received signal quality as well as traffic capacity in these cells. **Advantages of frequency reuse:** • Large coverage area • Efficient spectrum utilization • Enhanced system capacity